Dylan Becchetti

Mechanical Engineering | Engineering Software & Automation

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Portfolio: formsby-systems.github.io/engineering-portfolio

PROFILE

Lehigh mechanical engineering graduate with three Cardinal Glass manufacturing internships and independent work in research software and CAD. Available January 2027 for full-time U.S. remote work.

EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Built an Excel/VBA tracker that combined separate R&D line files into a daily part-status view used in shift debriefs. Defined the view with engineers and operators.
- Wrote illustrated assembly instructions, planned spare-parts storage and rack signage across 14 locations, and measured cleanroom airflow with a hood and micromanometer.
- Inspected pilot-line equipment, analyzed data in Minitab, drafted AutoCAD layouts and contributed to design failure mode and effects analysis.

SELECTED PROJECTS

Midas Futures | Creator & Technical Lead

- Led a year of development on a Python research and Practice platform with modular strategies, reproducible batch replay, checkpointed ablations and Monte Carlo tests.
- Unified historical and forward rules, matching 228 session decisions and 225 trade lifecycles against a sealed reference. Reproduced 11 decision files byte for byte in a separate Railway run.
- Built a research dashboard that shows account fees and modeled cash separately from trading P&L. An installed version displayed 1,410 modeled cases across 705 strategy variants.

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Led design revisions and modeled a full-scale suspended sculpture in OpenSCAD. Widened the ring gear to 79.5 inches and moved the drive while retaining the inner tooth profile. CNC-cut a full-size gear and checked bore fit with the team.

Operalith | Founder & Product Lead, independent project

- Built a configurable Next.js workspace where users choose tools and features while keeping records connected. Designed account saves that prevent duplicate writes and reject stale edits. Verified that undo preserved 11 prior history entries.

Scanned Leverage and Optimized Production | Creator

- Built a Python pipeline that compiles different Roblox place files from the same hash-pinned Luau modules. Connected market discovery to reusable components and CI-checked build recipes.

Formsby | Creator, search prototype

- Designed a cross-platform request router and built the housing prototype with queued searches, listing verification and a private results dashboard.

EDUCATION & TOOLS

Lehigh University | B.S. Mechanical Engineering, May 2026

Materials Science & Engineering minor expected Dec. 2026 | SOLIDWORKS Associate (CSWA), 2024.

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Minitab.